

Chapter 18

Joint Distributions, Marginals, Conditionals

Lecture 18 | Devore 9e: 5.1–5.2

18.1 Warm-up: linear transformations

Example 18.1 (Curving an exam). Scores X_{old} are to be rescaled by $X_{\text{new}} = aX_{\text{old}} + b$ so that the top score of 100 stays 100 and the class average of 75 becomes 83 (a B). Solve

$$\begin{cases} 100a + b = 100 \\ 75a + b = 83 \end{cases} \implies 25a = 17, \quad a = 0.68, \quad b = 32.$$

Recall from Lectures 9–10 that this shifts the mean to $a\mu + b$ and *shrinks* the standard deviation to $a\sigma = 0.68\sigma$ — a curve of this kind compresses the class together as well as moving it up.

18.2 From one random variable to two

Everything so far concerned a single X . But the interesting questions are usually about relationships: height *and* weight, auto premium *and* home premium, almonds *and* cashews in the same can. So we now study pairs (X, Y) defined on the same sample space.

18.2.1 The discrete case

Definition 18.2 (Joint pmf). For discrete X, Y on the same sample space,

$$p(x, y) := \mathbb{P}\{X = x \text{ and } Y = y\},$$

which satisfies $p \geq 0$ and $\sum_x \sum_y p(x, y) = 1$. For any set A of pairs,

$$\mathbb{P}\{(X, Y) \in A\} = \sum_{(x, y) \in A} p(x, y).$$

Example 18.3 (Insurance). X is the auto deductible and Y the homeowner's deductible for a randomly chosen customer:

$p(x, y)$	$y = 500$	$y = 1000$	$y = 5000$	$p_X(x)$
$x = 100$	0.30	0.05	0	0.35
$x = 500$	0.15	0.20	0.05	0.40
$x = 1000$	0.10	0.10	0.05	0.25
$p_Y(y)$	0.55	0.35	0.10	1

Definition 18.4 (Marginal pmf).

$$p_X(x) = \sum_y p(x, y), \quad p_Y(y) = \sum_x p(x, y).$$

Marginals are the row and column sums — literally written in the margins of the table, which is where the name comes from. Note that the marginals do *not* determine the joint pmf: many tables have the same marginals.

18.2.2 The continuous case

Definition 18.5 (Joint pdf). $f(x, y)$ is a **joint pdf** if $f \geq 0$ and $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$; then

$$\mathbb{P}\{(X, Y) \in A\} = \iint_{(x, y) \in A} f(x, y) dx dy.$$

Where a single-variable pdf was a curve and probability was area under it, a joint pdf is a *surface* and probability is the volume under it over a region.

Definition 18.6 (Marginal pdf).

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy, \quad f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx.$$

Example 18.7. $f(x, y) = \frac{6}{5}(x + y^2)$ for $0 \leq x \leq 1$, $0 \leq y \leq 1$. First check that this is a density: $\int_0^1 \int_0^1 \frac{6}{5}(x + y^2) dy dx = \frac{6}{5}(\frac{1}{2} + \frac{1}{3}) = 1$. ✓ Then

$$\mathbb{P}\left\{0 \leq X \leq \frac{1}{4}, 0 \leq Y \leq \frac{1}{4}\right\} = \int_0^{1/4} \int_0^{1/4} \frac{6}{5}(x + y^2) dy dx = \frac{6}{5} \left(\frac{1}{128} + \frac{1}{768}\right) = \frac{7}{640} \approx 0.0109.$$

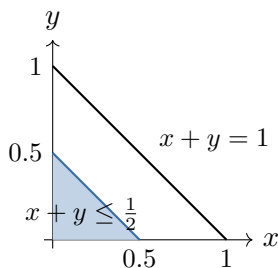
Example 18.8 (Can of nuts — the running example). A 1-lb can of “deluxe mixed nuts” contains almonds, cashews and peanuts. Let X be the weight of almonds and Y the weight of cashews, so $1 - X - Y$ is the weight of peanuts. Take

$$f(x, y) = 24xy \quad \text{for } x, y \geq 0, x + y \leq 1,$$

and 0 otherwise. Note the support is a *triangle*, not a rectangle — this is where most of the difficulty lives. It really is a density:

$$\int_0^1 \int_0^{1-x} 24xy \, dy \, dx = \int_0^1 12x(1-x)^2 \, dx = 1. \quad \checkmark$$

What is $\mathbb{P}\{X + Y \leq 0.5\}$? The region is the triangle below the line $y = -x + \frac{1}{2}$:



$$\mathbb{P}\{X + Y \leq 0.5\} = \int_0^{1/2} \left(\int_0^{-x+1/2} 24xy \, dy \right) dx = \int_0^{1/2} 12x \left(\frac{1}{2} - x \right)^2 dx = \frac{1}{16} = 0.0625.$$

Remark 18.9. The hard part of every joint-pdf problem is the *limits of integration*, not the integrand. Draw the region first. Always.

18.3 Independence

Definition 18.10 (Independent random variables). X and Y are **independent** if

$$(\text{discrete}) \quad p(x, y) = p_X(x) p_Y(y) \quad \text{or} \quad (\text{continuous}) \quad f(x, y) = f_X(x) f_Y(y)$$

for all x, y . More generally $X_1, \dots, X_n$ are independent if every subcollection has joint pmf/pdf equal to the product of its marginals.

Example 18.11. $X \sim \text{Exponential}(\lambda_1)$ and $Y \sim \text{Exponential}(\lambda_2)$ independent gives

$$f(x, y) = \lambda_1 e^{-\lambda_1 x} \cdot \lambda_2 e^{-\lambda_2 y} = \lambda_1 \lambda_2 e^{-(\lambda_1 x + \lambda_2 y)}, \quad x, y \geq 0,$$

so, for instance, $\mathbb{P}\{X \geq 1500, Y \geq 1500\} = e^{-1500\lambda_1} e^{-1500\lambda_2}$.

Example 18.12 (Which fails first?). Return to the two bulbs of Lecture 15: independent exponential lifetimes X and Y with rates $\lambda_1 = \frac{1}{1000}$ and $\lambda_2 = \frac{1}{500}$ per hour. What is the probability that the bulb with mean lifetime 1000 hours fails first?

Integrate the joint density over the region $\{x < y\}$:

$$\mathbb{P}\{X < Y\} = \int_0^\infty \int_x^\infty \lambda_1 \lambda_2 e^{-\lambda_1 x - \lambda_2 y} \, dy \, dx = \int_0^\infty \lambda_1 e^{-\lambda_1 x} e^{-\lambda_2 x} \, dx = \frac{\lambda_1}{\lambda_1 + \lambda_2}.$$

Here that is $\frac{1/1000}{3/1000} = \frac{1}{3}$. Each bulb's chance of going first is its share of the total rate. And by memorylessness, once one bulb has failed, the survivor is as good as new, so a race among several exponential clocks can be followed one failure at a time.

Remark 18.13 (A fast test for dependence). In the can-of-nuts example $f(x, y) = 24xy$ looks like a product — but the support $\{x + y \leq 1\}$ is not a rectangle, so knowing $X = 0.9$ forces $Y \leq 0.1$. The variables are dependent. Rule of thumb: if the support is not a product region, the variables cannot be independent no matter how the formula factors.

Example 18.14 (Maxima and minima). If $X_1, \dots, X_n$ are independent, then for $T_{\max} = \max(X_1, \dots, X_n)$,

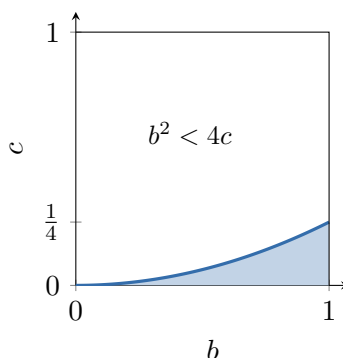
$$\mathbb{P}\{T_{\max} \leq t\} = \mathbb{P}\{X_1 \leq t, \dots, X_n \leq t\} = \prod_{i=1}^n F_{X_i}(t),$$

and similarly $\mathbb{P}\{T_{\min} > t\} = \prod_i (1 - F_{X_i}(t))$. This is the standard way to handle “the first component to fail” and “the last one standing”.

Example 18.15 (A random quadratic). B and C are independent and uniform on $[0, 1]$. What is the probability that $x^2 + Bx + C = 0$ has real roots?

The roots are real when the discriminant is non-negative, $B^2 \geq 4C$. The joint density is 1 on the unit square, so the probability is the area of the shaded region below the curve $c = b^2/4$:

$$\mathbb{P}\{B^2 \geq 4C\} = \int_0^1 \frac{b^2}{4} db = \frac{1}{12}.$$



If instead B is uniform on $[0, 2]$, the joint density is $\frac{1}{2}$ on $[0, 2] \times [0, 1]$, the same kind of region now has area $\int_0^2 \frac{b^2}{4} db = \frac{2}{3}$, and the probability rises to $\frac{1}{3}$.

18.4 Conditional distributions

Definition 18.16 (Conditional pdf/pmf). For $f_X(x) > 0$,

$$f_{Y|X}(y | x) := \frac{f(x, y)}{f_X(x)}, \quad \text{and symmetrically} \quad f_{X|Y}(x | y) := \frac{f(x, y)}{f_Y(y)}.$$

The discrete version replaces f by p throughout.

Compare with $\mathbb{P}(A | B) = \mathbb{P}(A \cap B)/\mathbb{P}(B)$ from Lecture 5: same shape, with the joint density playing the role of $\mathbb{P}(A \cap B)$ and the marginal that of $\mathbb{P}(B)$.

For each fixed x , $f_{Y|X}(\cdot | x)$ is a genuine pdf in y , so it can be integrated like any other:

$$\mathbb{P}\{Y \leq 0.5 | X = 0.2\} = \int_{-\infty}^{0.5} f_{Y|X}(y | 0.2) dy.$$

Remark 18.17. Conditioning on $\{X = 0.2\}$ is conditioning on an event of probability zero, which Lecture 5 did not allow. The definition above is how that is made legitimate: it is a limit of conditioning on $\{0.2 - \delta \leq X \leq 0.2 + \delta\}$.

Proposition 18.18. *X and Y are independent if and only if $f_{Y|X} = f_Y$ and $f_{X|Y} = f_X$; that is, conditioning changes nothing.*

Example 18.19 (Can of nuts, conditionals). With $f(x, y) = 24xy$ on the triangle,

$$f_X(x) = \int_0^{1-x} 24xy \, dy = 12x(1-x)^2, \quad 0 \leq x \leq 1,$$

so for $0 < x < 1$,

$$f_{Y|X}(y | x) = \frac{24xy}{12x(1-x)^2} = \frac{2y}{(1-x)^2}, \quad 0 \leq y \leq 1-x.$$

Given that the can holds x pounds of almonds, the cashew weight is Beta(2, 1)-shaped on $[0, 1-x]$ — and the range depends on x , which is dependence made visible.